

Sai Teja Dusari

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🌐 [LinkedIn](#)

🐙 [GitHub](#)

📁 [Portfolio](#)

🎓 [Google Scholar](#)

Education

Columbia University

New York, NY

Master of Science in Data Science

Expected Dec 2026

Coursework: Computer Systems for Data Science, Applied Machine Learning, Elements of Data Science, Statistical Inference and Modeling, Exploratory Data Analysis and Visualization, Generative AI using LLM's.

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology in Artificial Intelligence (CGPA: 4.00/4.00)

Sep 2021 – Aug 2025

Coursework: Data Structures and Algorithms, Neural Networks and Machine Learning, Database Management Systems, Probability and Statistics, Deep Learning Techniques, Foundation of Data Analysis, NLP, Cloud Computing, Operating System Design.

Professional Experience

STARR INSURANCE – AI Intern – New York, NY

Jun 2026 – Aug 2026

- Developed a **Claims Similarity Search** tool using **TF-IDF** and embedding-based retrieval to identify relevant historical claims from unstructured text descriptions.
- Developed multiple machine-learning models to predict **claim severity** throughout the claim lifecycle, achieving **90%** accuracy on two-stage **Gaussian model**.
- Presented methodology, **prototype**, and recommendations to the **AI/ML team** every week, incorporating technical and stakeholder feedback.
- Built an AI-powered data quality pipeline in Python that automatically profiles datasets, detects anomalies, and generates prioritized business rules using **Meta LLaMA 3.3 8B**, replacing a manual multi-step process.

Research Experience

Research Assistant – SRMIST

Mar 2024 – Jan 2025

HVD-Random Forest Classifier-based Domain Adapted Multi-Class Arrhythmia Recognition

[Link](#)

- Initiated and led a 3-member research team, defining project scope in collaboration with faculty, structuring milestones, and driving weekly discussions to ensure steady progress and alignment with research objectives.
- Applied **Hilbert Vibration Decomposition (HVD)** for extracting intrinsic mode functions and leveraged **CORAL Domain Adaptation** to address distribution shifts between source and target datasets, improving generalization.
- Optimized a Domain-Adapted Random Forest Classifier through hyperparameter tuning and feature selection, achieving **98%** test accuracy and improved cross-domain performance.

CBAM-Based Hybrid CNN-Autoencoder Model for Multiclass Classification of Brain Tumor Images

[Link](#)

- Proposed and led project direction, coordinating task allocation, guiding implementation strategies, and mentoring team members on deep learning concepts.
- Architected hybrid CNN-Autoencoder models integrated with attention mechanisms to capture spatial features and improve representation learning for multiclass classification.
- Attained **95%** test accuracy on **1,400** brain tumor images by incorporating the Convolutional Block Attention Module (CBAM), enhancing feature focus and reducing misclassification.
- Co-authored and presented findings in a conference paper, contributing to experimental design, result analysis, and technical writing while incorporating peer and reviewer feedback.

Academic Projects

ResumeFit: AI Semantic Matching for Jobs

Oct 2025 – Dec 2025

- Built a **FastAPI + SQLite** AI job platform that auto-applied to **75%+** matching roles using **sentence-transformer** embeddings and hybrid scoring, reducing application time by **80%**.
- Designed a resume optimization engine generating improved resumes for **65–75%** matches by auto-injecting **10–15** missing keywords, achieving **under 2s** analysis per job using **TF-IDF** vectorization.
- Deployed a secure multi-user system with authentication, profile management, and automated workflows tracking **100+** applications across **30+** pre-loaded tech job descriptions.

[Link](#)

Smart Diet Recommendation System with LLM Integration

Nov 2025 – Dec 2025

- Architected ML nutrition platform with k-NN engine processing **500K+ recipes**, achieving **95% budget accuracy** and **156ms query latency** via cosine similarity optimization.
- Engineered microservices system using FastAPI + Streamlit, deployed to **Hugging Face Spaces** with Docker, supporting 3 modules with RESTful API architecture.
- Pioneered budget-aware algorithm integrating **TinyLlama-1.1B LLM** for AI meal planning, automated shopping lists, and cost estimation with 2.30 / day variance.

[Link](#)

Technical Skills

Programming Languages: Python, SQL, R, C/C++, MATLAB, HTML, CSS, JavaScript.

Data Analysis & Visualization: Excel, Tableau, Power BI, Matplotlib, Seaborn.

Machine Learning & Libraries: Scikit-learn, NumPy, Pandas, NLTK, spaCy, Transformers, OpenCV, Torchvision.

Frameworks & Gen AI: TensorFlow, Keras, PyTorch, Streamlit, PySpark, LangChain, LlamaIndex, RAG, Prompt Engineering.

Cloud & Platforms: AWS, Microsoft Azure, Google Cloud Platform (GCP).

Tools and Data Platforms: Snowflake, Databricks, Git/GitHub, Docker, ETL Pipelines, A/B Testing.